ALGEBRAIC STRUCTURES

Introduction

Let A be any set. A Mapping $f : A \times A \ldots\ldots \times A \to A$ (or) $f : A^n \to A$ is called an n-ary operation and 'n' called the order of the operation.

Unary operation:

$n=1,\ f : A \to A$

Binary operation:

$n=2,\ f : A \times A \to A$

Algebraic system (or) algebra:

A set together with a number of operations (binary) on the set is called an **Algebraic system**

Example

- Semi groups
- Monoids
- Groups are algebraic systems with one binary operation.
- Rings, Integral domains ,Fields are algebraic systems with two binary operation.

Properties of Binary operations

- **Closure property:** A binary operation $*$: $\mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$ is said to be closed if for all a, b Ɛ G an element an element $a*b = x$ Ɛ G.
- **Associative:** $\mathbf{a*(b*c) = (a*b)*c}$, for all a, b, c Ɛ G
- **Existence of Identity:** There exists an element e Ɛ G such that

  $e*a = a*e = a$, a Ɛ G for all. The elements e called the identity element.
- **Existence of inverse:** For a ƐG, there exists an element b Ɛ G such that That $a*b = b*a = e$. The element b is called the inverse of 'a' and it is denoted by $b = a^{-1}$
- **Commutative:** For all a, b Ɛ G, if $a*b = b*a$, then* is commutative(abelian)
- **Distributive properties:**

  $a*(b . c) = (a*b) . (a*c)$ (**left distributive law**)

  $(b . c)*a = (b*a) . (c*a)$ (**right distributive law**)

  for all a, b, c Ɛ G
- **Cancellation properties:**

  $a*b = a*c \Rightarrow b = c$ **[left cancellation Law]**

  $b*a = c*a \Rightarrow b = c$, for all a, b, c Ɛ G [**right cancellation Law**]

Semi groups:

- **Definition:**

  A non-empty set S together with the binary operation *: **S X S → S** is said to be a **Semi group** if * satisfies the following conditions, namely the closure property and associative property. We denote the semi group by **(S,*)**.

- **Example.1**

  Let N={1, 2, 3, ……} be the set of natural numbers. Then **(N, +)** and **(N, X)** are semi group under the binary operations addition and multiplication respectively.

- **Example.2**

  Let E={2, 4, 6, ……} be the set of all even numbers. Then **(E, +)** and **(E, X)** are semi groups.

- **Morphism of semi groups:**

  Let **(S, *)** and **(T, Δ)** be any two semi groups. A mapping f : S→T is said to be a **semi group homomorphism** if f f(a*b) = f(a) Δ f(b), for all a, b ε S.

$f : (S, *) \xrightarrow[\text{1-1}]{\text{homomorphism}} (T, \Delta)$ → monomorphism

$f : (S, *) \xrightarrow[\text{Onto}]{\text{homomorphism}} (T, \Delta)$ → epimorphism

$f : (S, *) \xrightarrow[\text{1-1 and onto}]{\text{homomorphism}} (T, \Delta)$ → isomorphism

$f : (S, *) \xrightarrow[\text{1-1}]{\text{isomorphism}} (T, \Delta)$ → endomorphism

$f : (S, *) \xrightarrow[\text{onto}]{\text{isomorphism}} (T, \Delta)$ → automorphism

Example:

Let (N, +) and (Z_e, $+_m$) be any two semi groups.

Define a map f : N→Z_m by f(a) = $[a]_m$, for all a ε N.

Then f(a + b) = $[a + b]_m$ = [a] $+_m$ [b] = f(a) $+_m$ f(b)

Therefore f is a semi group homomorphism.

Monoids:

- **Definition:**

  A semi group **(M, *)** with an identity element with respect to the operation * is called a **Monoid (or)** a non empty set M together with the binary operation $*: M \times M \rightarrow M$ is said to be a **Monoid if *** satisfies the closure property, associative property and identity property.

- **Example.1**

  Let N= {1, 2, 3, ……} be the set of natural numbers. Then **(N, X)** is a monoid with the identity element **1**, but **(N, +)** is not a monoid since the additive identity **0** is not a natural number.

- **Example.2**

  Let Z_+ **= {0, 1, 2, 3, ……}** be the set of all non-negative integers. Then (Z_+, +) and (Z_+, X) are semi groups as well as monoids.

- **Example.3**

  Let S be a nonempty set and P(s) be its power set. Then (P(s), U) and (P(s), ∩) are monoids with identities $\boldsymbol{\varphi}$ and S respectively.

- **Morphism of monoids:**

Let $(M, *, e_M)$ and (T, Δ, e_T) be any two monoids with the identity elements e_M and e_T respectively. A mapping

$f : M \to T$ is said to be a **monoid morphism** if for any two elements a, b, ε M, $f(a*b) = f(a)\ \Delta\ f(b)$ and $f(e_M) = e_T$

$$f : (M, *, e_M) \xrightarrow{1-1} (T, \Delta, e_T) \qquad \to \text{monomorphism}$$

$$f : (M, *, e_M) \xrightarrow{onto} (T, \Delta, e_T) \qquad \to \text{epimorphism}$$

$$f : (M, *, e_M) \xrightarrow{\text{1-1 and onto}} (T, \Delta, e_T) \qquad \to \text{isomorphism}$$

- **Theorem 1:**

For any commutative monoid (M, *), the set of idempotent element of M forms a submonoid.

Proof:

Let (M, *) be a commutative monoid.

Let S = {a ε M/a*a=a}, the set of idempotent elements of M.

Clearly e ε S, as e x e = e.

Let a, b ε S with a * a = a b * b = b.

Now (a * b) * (a * b) = (a *b) * (b * a)

= a * (b * b) * a

= a * (b) * a

= a * (b * a)

= a * (a * b)

= (a * a) * b

= a * b

Hence a * b ε S

Therefore (S, *) is a sub monoid of (M, *)

GROUPS

Definition (1):

A non-empty set G together with the binary operation *, (G, *) is called a **group** if * satisfies the following conditions

- **Associative:** For every a, b, c Ɛ G, a*(b*c)=(a*b)*c .
- **Identity:** There exists an element e ƐG called the identity element such that a*e=e*a=a, for all a Ɛ G.
- **Inverse:** There exists an element a^{-1} Ɛ G called the inverse of 'a' such that $a* a^{-1}= a^{-1} *a=e$, for each a Ɛ G

Definition (2):

A group (G, *) is said to be abelian group if a*b=b*a, for all a, b Ɛ G. otherwise it is non-abelian group.

Definition (3):

The order of a group (G, *), denote by O(G) or $|G|$, is the number of elements of G, is finite.

Example (1):

The set of all integers Z with the operation of ordinary addition is an infinite abelian group.

Example (2):

The set of all integers Z is not a group under multiplication. (Z, .) is not a group. Because there is no multplicaticative inverse in Z. But (Z, .) is a monoid and hence a semi group.

Example (3):

The set of all non-zero real numbers R from an infinite abelian group under the binary operation * defined by a*b=ab/2 , for all a, b ƐR

Example (4):

The set of all non-zero real numbers R, under the operation of ordinary multiplication is a group. An inverse of $a\neq 0$ is 1/a i.e. $a^{-1} = 1/a$

Example (5):

Prove that the set $A=\{1,\omega,\omega^2\}$ is an abelian group of order 3 under multiplication. Where 1, ω, ω^2 are cube roots of unity and $\omega^3=1$.

Properties of groups

Properties of groups

Property 1:

The identity of a group is unique (or) If (G *) is a group and *e* is an identity of G, then no other element of G is an identity of G.

Proof:

Suppose that e_1 and e_2 are two identities of the group (G, *).

Now e_1 is the identity, then $e_{1*}\, e_2 = e_{2*}\, e_1 = e_2$ $\rightarrow$ (1)

Again e_2 is the identity, then $e_{2*}\, e_1 = e_{1*}\, e_2 = e_1$ $\rightarrow$ (1)

From (1) and (2) we see that $e_1 = e_2$

The identity is unique

Property 2:

In a group , the left and right cancellation laws are true. That $a*b=a*c \Rightarrow b=c$ and $b*a=c*a \Rightarrow b=c$.

Proof:

Let $G=(G, *)$ be a group For $a \mathcal{E} G$, there exists an $a^{-1} \mathcal{E} G$ such that $a* a^{-1}= a^{-1}*a=e$.

(i)Left cancellation law:

Let $a*b=a*c$.

$a^{-1}*(a*b)= a^{-1}*(a*c) \Rightarrow(a^{-1}*a)*b=(a^{-1}*a)*c$

$\Rightarrow e*b=e*c \Rightarrow b=c$.

(ii)Right cancellation law :

$\Rightarrow$ Let $b*a =c*a \Rightarrow (b*a)* a^{-1}=(c*a)* a^{-1}$

$\Rightarrow b*(a* a^{-1}) =c*(a*a^{-1})$

$\Rightarrow b*e= c*e \quad \Rightarrow b=c$.

Property 3:

The inverse of any element of group is unique.

Proof:

Let G be a group, a$\mathcal{E}$ G and e be the identity element of G.

Suppose that b and c be two inverses of an element a$\mathcal{E}$ G.

$b * a = e = a * b$ and $c * a = e = a * c$.

Now $(b*a) * c = c \Rightarrow b*(a*c) = c$

$\Rightarrow b * e = c \Rightarrow b = c$

There is one and only one inverse for each element in a group.

Property 4:

A group can not have any element which is idempotent expect the identity element

Proof:

Let G be a group. Let us assume that $a \in G$ is idempotent. Then

$a*a = a.$

Now $e = a^{-1} * a = a^{-1}*(a*a) = (a^{-1}*a)*a$

$= e*a = a$

$\Rightarrow e = a$

Property 5: If a is an element of group G, then $(a^{-1})^{-1} = a$ (or) If the inverse of a is a^{-1}, then the inverse of a^{-1} is a.

Proof:

If e is the identity of the group G, then

$a^{-1}*a = e = a*a^{-1}$, for every a Ɛ G.

$(a^{-1})^{-1} * (a^{-1}*a) = (a^{-1})^{-1} * e = (a^{-1})^{-1}$

$\Rightarrow (a^{-1})^{-1} * (a^{-1}*a) = (a^{-1})^{-1}$

$\Rightarrow ((a^{-1})^{-1} * a^{-1})*a = (a^{-1})^{-1}$

$\Rightarrow e*a = (a^{-1})^{-1}$

$\Rightarrow a = (a^{-1})^{-1}$

Property 6:

If a has inverse b and b has inverse c, then a=c.

Proof:

Given $a*b = e = b*a \quad \rightarrow (1)$

and $\quad b*c= e = c*b \quad \rightarrow (2)$

$$a = a*e = a*(b*c)$$

$$= e*c$$

$$= c$$

$$a = c$$

Property 7:

The inverse of the product of two element of a group is the product of their inverse order. That is $(a * b)^{-1} = b^{-1} * a^{-1}$.

Proof:

Let a, b Ɛ G and a^{-1} , b^{-1} be their inverse respectively .

$a * a^{-1} = e = a^{-1} * a$ and $b * b^{-1} = e = b^{-1} * b$

Now $(a*b)*(b^{-1}*a^{-1}) = a*[b*(b^{-1}*a^{-1})]$

$= a*[(b*b^{-1})*a^{-1}]$

$= a*[e*a^{-1}] = a*a^{-1}$

$= e$

Similarly we can prove $(b^{-1}*a^{-1})*(a*b) = e$.

$(a*b)*(b^{-1}*a^{-1}) = e = (b^{-1}*a^{-1})*(a*b)$.

We see that $(b^{-1}*a^{-1})$ is the inverse of $(a*b)$

That is $(a * b)^{-1}$ is the inverse of $(a*b)$

That is $(a * b)^{-1} = b^{-1}*a^{-1}$

Property 8:

There existence of a unique inverse of every element of G guarantees the unique solvability of any equation of the type $a*x = b$ has a unique solution $x = a^{-1}*b$. in addition, the equation $x*a=b$ has a unique solution $x = b* a^{-1}$.

Proof:

Let G be a group. The for any a, b ƐG,

we have $a* a^{-1} = e = a^{-1}*a$ and $b*b^{-1} = e = b^{-1}*b$.

Now $a * x = b = e * b$

$= (a*a^{-1})*b$

$= a * (a^{-1}*b)$

$\Rightarrow\ x = a^{-1}*b$

To prove this solution is unique, let us suppose x_1 and x_2 are two solutions of $a* x = b$.

Then

$a * x_1 = b$ and $a*x_2 =b$

$a* x_1 = a* x_2$

$\Rightarrow\ x_1 = x_2$

$x = a^{-1}* b$ is the unique solution for $a* x = b$. similarly we can prove the second part.

Property 9:

A group G is abelian iff $(a * b)^2 = a^2 * b^2$

Proof:

Let us assume that G is abelian. Hence, for a, b Ɛ G.

We have $a * b = b * a$

Now

$$
\begin{aligned}
a^2 * b^2 &= (a * a) * (b * b) \\
&= a * [a * (b * b)] \\
&= a * [(a * b) * b] \\
&= a * [(b * a) * b] \\
&= (a * b) * (a * b) \\
&= (a * b)^2
\end{aligned}
$$

$$(a * b)^2 = a^2 * b^2$$

Conversely, assume that $(a * b)^2 = a^2 * b^2$

$\Rightarrow (a * b) * (a * b) = (a * a) * (b * b)$

$\Rightarrow a * [b * (a * b)] = a * [a * (b * b)]$

$\Rightarrow b * (a * b) = a * (b * b)$

$\Rightarrow (b * a) * b = (a * b) * b$

$\Rightarrow b * a = a * b$

$\Rightarrow$ G is abelian

Property 10:

For any group G, if $a^2 = e$ with $a \neq e$, then G is abelian

(0r)

If every element of group G is its own inverse, then G is abelian. Is the converse also true.

Proof:

Let G be a group and a, b ƐG, then a * b ƐG.

Given $a = a^{-1}$ and $b = b^{-1}$

$$
\begin{aligned}
(a * b) &= (a * b)^{-1} \\
&= b^{-1} * a^{-1} \\
&= b * a
\end{aligned}
$$

$\Rightarrow$ G is abelian

Modular Arithmetic

Modular Arithmetic:

- The addition modulo *n* is defined by a $+_n$ b =r ,where a, b are integers, *n* is a +ve integer and *r* is the least +ve reminder when (a+b) is divided by *n*.

a $+_n$ b =the least +ve reminder when (a+b) is divided by *n*.

Eg: $8 +_5 3 = 11 = 2(5)+1 = 1$ (Addition mod 5)

$-23 +_7 3 = -20 = -3(7) + 1 = 1$ (Addition mod 7)

The multiplication modulo *n* is defined by a b = the least +ve reminder when (a b) is divided by *n.*

3 5 = 15 = 3(4) + 3 =3

-5 6 = -30 = -8(4) + 2 = 2

Properties of Modular Arithmetic on Z_n

Addition modulo *n* is always commutative and associative. 0 is the identity for $+_n$ and every element of Z_n has an additive inverse .

Multiplication modulo *n* is always commutative and associative and 1 is identity for $_n$. multiplication modulo *n* is distributive over addition modulo *n* . also every element of Z_n has the multiplicative inverse.

Example 1:

Prove that $(Z_3, +_3)$ is an abelian group.

Let Z_3 ={0,1,2}, Let us defined the addition table as

$+3$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

For $x, y \in Z_{3,}$ define,

$x +_3 y$ = the least +ve reminder , when $(x+y)$ divided by 3,

since all the entries in the addition table are the element of Z_3, it is closed under $+_3$ .Associative, commutative, with identity element 0 and every element of Z_n has an additive inverse , we see that $(Z_3, +_3)$ is an additive abelian group.

Example 2:

Prove that $(Z_5, \times_5)$ is an abelian group.

Solution:

Let $Z_5 = \{1,2,3,4\}$. Let us define the multiplication mod 5 table as

$\times_5$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Since all the entries in the multiplication table are the element of $Z_{5,}$
It is closed under $\times_5$. Here 1 is the identity. Also from fact that multiplication mod *n* is always associative, commutative and every element of Z_n has a multiplicative inverse, we see that $(Z_5,\times_5)$ is an abelian group.

Sub Groups

Definition:

A non-empty subset H of a group G is said to be a subgroup of G if H itself is a group under the same operation defined on G and with the same identity element.

Let (G, *) be a group and H G ($H \neq \varphi$) is said to be subgroup (H, *) of (G, *) if

1. $e \in H$, where e is the identity of G
2. For any $a \in H$, $a^{-1} \in H$
3. For $a, b \in H$, $a * b \in H$.

Example :

1. The set of all integers is a subgroup of the set of all real numbers under addition. That is (Z, +) is a subgroup of (R, +)
2. The set of even integer $2Z = \{2k / k \in Z\}$ is a subgroup of (Z, +)
3. The set of non-negative integers is not a subgroup of (Z, +), because, expect 0, no element have the additive inverse.

Theorem 1

The necessary and sufficient condition that a non-empty subset H of a group G be a subgroup is $a \in H, b \in H \Rightarrow a * b^{-1} \in H$

Proof

Necessary Conditions:

Assume that H is a subgroup of G. since H itself is a group ; we have $a, b \in H$.
Also $b \in H \Rightarrow b^{-1} \in H$.
$a, b \in H$ $a, b^{-1} \in H \Rightarrow a * b^{-1} \in H$

Sufficient Conditions : Let $a * b^{-1} \in H \rightarrow (1)$ for all $a, b \in H$ and H G. we have to prove that H is a subgroup of G.

For,

Identity : Let $a \in H, a \in H \Rightarrow a * a^{-1} \in H$

$\Rightarrow e \in H$

Hence the identity e is the element of H

Existence of inverse :

Let $e \in H, a \in H \Rightarrow e * a^{-1} \in H \Rightarrow a^{-1} \in H$

Every element of H has an inverse which is in H.

•Closure :

Let $b \in H \Rightarrow b^{-1} \in H$

For $a, b \in H \Rightarrow a, b^{-1} \in H \Rightarrow a * (b^{-1})^{-1} \in H$

$\Rightarrow a * b \in H$

H is closed under the composition.

Associative :

Since H   G, the elements of H are also the elements of G. since the composition in G is associative, it must also be associative in H.

H itself is a group for the composition in G.

H is a subgroup of G.

Theorem 2:

The intersection of two subgroup of a group is also a subgroup of the group (OR) If H_1, H_2 are two subgroup of a group G, then $H_1 \cap H_2$ is also a subgroup of G.

Proof:

Let H_1 and H_2 be any two subgroups of G then $H_1 \cap H_2 \neq \varphi$, because at least the identity element is common to both H_1 and $H_{2.}$

To prove $H_1 \cap H_2$ is a subgroup , it is enough if we show that, for a, b $\in H_1 \cap H_2 \Rightarrow a * b^{-1} \in H_1 \cap H_2 \Rightarrow a * b^{-1} \in H_1 \cap H_2$.

For let a, b $\in H_1 \cap H_2 \Rightarrow$ a, b $\in H_1$ and a, b $\in H_2$

$\Rightarrow a * b^{-1} \in H_1$ and $a * b^{-1} \in H_2$

$\Rightarrow a * b^{-1} \in H_1 \cap H_2$

a, b $\in H_1 \cap H_2 \Rightarrow a * b^{-1} \in H_1 \cap H_2$

$\Rightarrow H_1 \cap H_2$ is a subgroup of G

Theorem 3:

The union of two subgroup of G need not be a subgroup of G.

Proof:

To prove these consider the following example. Let G be a group of integers under addition

(Z, +) = (G, +) is group

Now define $H_1 = \{ x \mid x = 2n , n \in Z \} = \{0,\pm 2,\pm 4,\pm 6,\ldots\}$

$H_2 = \{ x \mid x = 3n , n \in Z \} = \{0,\pm 3,\pm 6,\pm 9,\pm 12\ldots\}$

We see that H_1 and H_2 are subgroup of G.

Define $H_1 \cup H_2 = \{ x \mid x \in H_1 \text{ or } x \in H_2 \}$

$= \{0,\pm 2,\pm 3,\pm 4,\pm 6,\pm 8,\pm 9,\ldots\ldots\}$

For 2,9 $H_1 \cup H_2 \Rightarrow 2+9 = 11 \ H_1 \cup H_2$

$H_1 \cup H_2$ is not closed under addition

$H_1 \cup H_2$ is not a group

$H_1 \cup H_2$ is not a subgroup of G

Cyclic Groups

Definition:

A group (G, *) is said to be cyclic if there exists a∈ G such that any $x \in G$ can be written as either $x = a^n$ or $x = na$, where n is some integer. Here the element 'a' is called the 'generator' of the cyclic group G. That is the cyclic group generated by 'a' and we denote if by G = (a)

Theorem 1:

Every cyclic group is abelian.

Proof:

Let (G, *) be a cyclic group generated by an element a∈ G. i.e G = (a). Then for any two elements $x, y \in G$, we have $x = a^n$, $y = a^m$, where m, n are integers.

$$\begin{aligned} x * y &= a^n * a^m = a^{n+m} = a^{m+n} \\ &= a^m * a^n \\ &= y * x \end{aligned}$$

(G, *) is abelian.

Theorem 2:

If 'a' is a generator of a cyclic group G, then a^{-1} is also a generator of G (or). If a is an element of group G, then $(a) = (a^{-1})$.

Proof

Let G = (a) be a cyclic group generated by 'a'. Then $a^r \in G$, where r is some integer we can write $a^r = (a^{-1})^{-r}$, since –r is also some integer

Each element of G is generated by a^{-1}

a^{-1} is also a generator of G.

$(G) = (a) = (a^{-1})$.

Theorem 3:

Prove that the group G = {1,-1,i,-i} is cyclic and find its generators.

Solution

We can write $1 = (i)^4$, $-1= (i)^2$, $i= (i)^1$ $–i = (i)^3$

i.e , All the elements of G can be expressed as integral powers of the element i.

G is a cyclic group generated by i.

Since 'i' is the generator of G, $(i)^{-1}$ is also a generator of G . Hence G is a cyclic group and its generators are I and $(i)^{-1} =1/i = -i$

Theorem 4:
Every subgroup of a cyclic group is cyclic

Theorem 5:
Let (G, *) be a finite group generated by an element $a \in G$. If G is of order n, that is $O(G) = n$, then $a^n = e$ so that $G = \{a, a^2, \ldots a^n = e\}$. Further, n is the least positive integer for which $a^n = e$.

Theorem 6:
The direct product of two or more groups is again a group.

Morphism of Group

Definition:

Let (G, *) and (H, Δ) be any two groups. A mapping f: G → H is said to be a **homomorphism** if $f(a * b) = f(a)\,\Delta\, f(b)$, for any a, b∈ G.

Theorem 1:

If f is a homomorphism of a group G into a group G′ , then

(i) Group homomorphism preserves identities

i.e., $f(e) = e'$, where e is the identity of G and e' is the identity of G′

(ii) (Group homomorphism preserves inverses

i.e., $f(a^{-1}) = [f(a)]^{-1}$, for all a∈ G.

Proof:

Let a ∈ G and f is a homomorphism from G into G' , then $f(a) \in G'$

$$f(a) * e' = f(a) = f(a * e)$$
$$= f(a) * f(e)$$
$$\Rightarrow e' = f(e)$$

Let a ∈ G, then $a^{-1} \in G$ and $a * a^{-1} = e = a^{-1} * a$

$$e' = f(e) = f(a * a^{-1})$$
$$= f(a) * f(a^{-1})$$
$$\Rightarrow f(a) * f(a^{-1}) = e'$$

We see that $f(a^{-1})$ is the inverse of $f(a)$ in G'

i.e., $[f(a)]^{-1} = f(a^{-1})$

Theorem 2:

Let f *be* a homomorphism from G to G'. Let f (G) be the homomorphic image of G into G' is a subgroup of G'.

Proof:

Let $f(G) = \{ f(x) / x \in G \}$

Clearly f (G) is a non-empty subset of G'.

Now for any $a', b' \in f(G)$, we have $f(a) = a'$, $f(b) = b'$, for $a, b \in G$.

$$\begin{aligned} a' * (b')^{-1} &= f(a) * f[b]^{-1} \\ &= f(a) * f(b^{-1}) \\ &= f(a * b^{-1}) \end{aligned}$$

But $a * b^{-1} \in G \Rightarrow f(a * b^{-1}) \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$

i.e., For $a', b' \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$

$\Rightarrow f$ (G) is a subgroup of G'.

Theorem 3:

Let $f : G \to G'$ be a group homomorphism and H is a subgroup of G' , then $f^{-1}(H)$ is a subgroup of G.

Proof:

Let $f^{-1}(H) = \{ x = f^{-1}(y) \in G \mid f(x) = y \in H \}$

Since H is a subgroup of G', the set $f^{-1}(H)$ will be a non-empty subset of G.

Now consider $x_1 = f^{-1}(y_1), x_2 = f^{-1}(y_2) \in f^{-1}(H)$ for any $y_1, y_2 \in H$

With $f(x_1) = y_1$ *and* $f(x_2) = y_2$.

$$\begin{aligned}
\text{Let } x_1, x_2 \in f^{-1}(H) &\Rightarrow f(x_1), f(x_2) \in H \\
&\Rightarrow f(x_1) * [f(x_2)]^{-1} \in H \\
&\Rightarrow f(x_1) * f(x_2^{-1}) \in H \\
&\Rightarrow f(x_1 * x_2^{-1}) \in H \\
&\Rightarrow x_1 * x_2^{-1} \in f^{-1}(H) \\
x_1, x_2 \in f^{-1}(H) &\Rightarrow x_1 * x_2^{-1} \in f^{-1}(H)
\end{aligned}$$

$f^{-1}(H)$ is a subgroup of G.

Kernel of a homomorphism:

Definition:

Let $f : G \rightarrow G'$ be a group homomorphism

The set of elements of G which are mapped into e', the identity of G' is called the kernel of f and is denoted by Ker(f)

i.e., $Ker(f) = \{ x \in G / f(x) = e'$,the identity of G' $\}$

Theorem:

Proof:

$Ker (f) = \{ x \in G / f(x) = e'$,the identity of G' $\}$

Since $f(e) = e'$ is true always, at least $e \in Ker (f)$

i.e., Ker(f) is a non-empty subset of G.

Let $a, b \in Ker (f)$, with $f(a) = e'$ and $f(b) = e'$

$$\begin{aligned} f(a * b^{-1}) &= f(a) * f(b^{-1}) \\ &= f(a) * (f(b))^{-1} \\ &= e' * e' \\ &= e' \end{aligned}$$

$\Rightarrow a * b^{-1} \in Ker (f)$

I.e., $a, b \in Ker (f) \Rightarrow a * b^{-1} \in Ker (f)$

Ker (f) is a subgroup of G.

Definition:

A mapping f from a Group G to a group G' is said to be an **Isomorphism** if

- f is a homomorphism i.e. $f(a \cdot b) = f(a) \cdot f(b)$, for any $a, b \in G$
- f is one-one. (injective) i.e. distinct elements of G have different f –image in G'.
- f is onto (surjective) i.e. every element of G' should be the f-image of some elements in G.

Cosets

Definition:

•For any $a \in G$, the set $a * H$ defined by $a * H = \{a * h / h \in H\}$ is called the **Left coset** of H in G determined by the element $a \in G$.

•For any $a \in G$, the set $H * a$ defined by

$H * a = \{ h * a / h \in H\}$ is called the **Right coset** of H in G determined by the element $a \in G$.

The element $a \in G$ is called the respresentative element of the left coset $a * H$ and right coset $H * a$

Note

The right or left coset of H in G is not empty.

Since $e \in H$, $e * H = H = H * e$, H itself is right as well as left coset .

$H * a$, $a * H$ are also subsets of G

If G is abelian, then $a * H = H * a$

In all the subsequence discussion in this chapter, aH means $a * H$ and Ha means $H * a$

Example.1

Let $G = \{ 1, a, a^2, a^3 \}$ be a group and $H = \{1, a^2\}$ is a subgroup of G under multiplication. Find all the cosets of H.

Solution:

Let us find the right cosets of H in G.

$H1 = \{ 1, a^2\} = H$

$Ha = \{a, a^2\}$,

$Ha^2 = \{a^2, a^4\} = \{a^2, 1\} = H$

$Ha^3 = \{a^3, a^5\} = \{a^3, a\} = Ha$

$H.1 = H = Ha^2 = \{1, a^2\}$ and $Ha = Ha^3 = \{a, a^3\}$

are two distinct right cosets of H in G. Similarly we can find the left cosets of H in G.

Example.2

Find the right cosets of {[0],[2]} in the group (Z_4, $+_4$).

Solution

Let Z_4 = {[0],[1],[2],[3]} be a group and H = {[0],[2] } be a subgroup of Z_4 under $+_4$ (addition mode 4).

The left cosets of H are

[0] + H = {[0],[2]} = H ;

[1] +H = {[1],[3]} ;

[2]+H = {[2],[4]} = {[2],[0]} = {[0],[2]} = H

[3]+H = {[3],[5]} = {[3],[1]} = {[1],[3]} =[1]+ H.

[0]+H = [2]+H = H and [1]+H = [3]+H are the two distinct left cosets of H in Z_4.

Note:

The union of all distinct left cosets of H in G is equal to G.

If H any subgroup of G and a ∈ H, then Ha = H = aH.

Theorem1:

If a ∈ Hb then Ha = Hb and if a ∈ Hb, then aH = bH.

Theorem2:

Any two right cosets of H in G are either disjoint or identical.

Definition:

A subgroup (H, *) of group (G, *) is said to be a **normal subgroup** of G, if for every $x \in$ G and for every h$\in$ H, $xhx^{-1} \in$ H xHx^{-1} H.

Theorem1:

A subgroup H of a group G is normal iff $xHx^{-1} = H$

Proof:

Let $xHx^{-1} = H \Rightarrow xHx^{-1}$ H, for all $x \in G$

$\Rightarrow$ H is a normal subgroup of G

Conversely,

assume that H is a normal subgroup of G, then xHx^{-1} H, for all $x \in G \rightarrow (1)$

Now $x \in G \Rightarrow x^{-1} \in G$, then

$x^{-1}H(x^{-1})^{-1}$ H, for all $x \in G$

$\Rightarrow x^{-1}Hx$ H $\Rightarrow x(x^{-1}Hx)\, x^{-1}$ xHx^{-1}

$\Rightarrow x^{-1}xHx\, x^{-1}$ $xHx^{-1} \Rightarrow$ e H e xHx^{-1}

$\Rightarrow$ H xHx^{-1}, for all $x \in G$ $\rightarrow (2)$

From (1) and (2)

$xHx^{-1} = H$, for all $x \in G$.

Definition:

Let (H, *) be a subgroup of the group (G, *). We define the congruency relation on G called a left cosets relation with respect to the subgroup H or a left cosets relation mod H denoted by + such that for a, b ∈ G, **a is congruent to b mod H** written a+b (mod H) iff $b^{-1} * a \in H$.

Theorem1:

The left cosets relation mode H on (G, *) defined by, for a, b ∈ G, a+b (mod H) iff $b^{-1} * a \in H$ is an equivalence relation.

Theorem2:

Let (H, *) be a subgroup of (G, *). The set of left cosets of H in G forms a partition of G.

Theorem3:

If (H, *) is a subgroup of a group (G, *) and right cosets of H in G, then there exists a one-to-one correspondence between the elements of H and Ha (or)

$$O(H) = O(Ha).$$

Note:

There is a one-to-one correspondence between H and aH.

$$O(H) = O(aH)$$

Lagrange's therom

Theorem1:

The order of each subgroup of a finite group is a divisor of the order of the group.

Proof:

Let (G, *) be finite group and O(G) = n. Let (H, *) be a subgroup of (G, *) and O(H) = m

Suppose that $h_1, h_2, h_3, \ldots\ldots h_m$ are the m members of H. For a ∈ G, the right coset Ha of H in G is defined by

$$Ha = \{h_1 * a, h_2 * a, h_3 * a, \ldots\ldots, h_m * a\}$$

Since there should be a one-to-one correspondence between H and Ha, the members of Ha are distinct.

Hence each right coset of H in G has m distinct members.

We know that any right cosets of H in G are either disjoint or identical. Since G is a finite group, the number of distinct right cosets of H in G will be finite, The union of these k distinct right cosets of H in G is equal to G.

Hence, if $Ha_1, Ha_2, Ha_3, \ldots, Ha_k$ are the k distinct right cosets of H in G, then

$$G = Ha_1 \cup Ha_2 \cup H(a_3 \cup \ldots\ldots\cup Ha_k$$

$$O(G) = O(Ha_1) + O(Ha_2) + O(Ha_3) + \ldots\ldots O(Ha_1)$$

$n = m + m + m + \ldots\ldots + m$

(k times)

$n = km \Rightarrow n/m = k \Rightarrow$ m is a divisor of n

$\Rightarrow$ O(H) is divisor of O(G)

$\Rightarrow$ O(H) divides O(G).

Hence proof.

Note

- The converse of the Lagrange thermo is not true. That is, if m is a divisor of n, then it is not necessary that G must have a subgroup of order m. For example, the alternating group A_4 of degree 4 is of order 12. But there is no subgroup of A_4 of order 6, though 6 is a divisor of 12
- It is evident from Lagrange's theorem that if G is a finite group of order n and if m is not a divisor of n, then can be no subgroup of G of order m.